DERIVED COMPLETENESS OF THE DERIVED LIMIT OF A SYSTEM KILLED BY POWERS OF THE ELEMENTS OF AN IDEAL

THE STACKS PROJECT AUTHORS

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Editorial note. The proof uses the macro R-lim, the bold L of the derived tensor product and the torsion functor T of the shared preamble, all of which the retained preamble supplies verbatim.

Editorial note. The two facts the proof invokes by name, namely that derived complete objects are closed under R-lim and that the vanishing of the torsion functor characterises derived completeness, are part of the framework established earlier in the same section; the counted proof states them as the reason for each reduction and contains no cross reference to a labelled item, so no navigation stub is needed.

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Allowed standard prerequisites (editorial).

- The category of derived complete objects of D(A) with respect to an ideal I is closed under the derived limit of an inverse system. invoked at proof lines 26297-26298 as a standing property of derived complete objects. Reduces the assertion for the derived limit K' to the assertion for each of the objects K tensor^L K_n.
- The vanishing of the derived torsion functor T(-, f) on an object is equivalent to f being locally nilpotent on that object, so an object killed by a power of f is derived complete with respect to the principal ideal generated by f. used at proof lines 26299-26301, where the assumption that f^e is zero on K tensor^L K_n is turned into T(K tensor^L K_n, f) = 0. Shows that each term of the system is derived complete with respect to I, one generator at a time, which finishes the proof after the reduction.

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Lemma 0.2. *Let A be a ring and let $I \subset A$ be an ideal. Let (K_n) be an inverse system of objects of $D(A)$ such that for all $f \in I$ and n there exists an $e = e(n, f)$ such that f^e is zero on K_n. Then for $K \in D(A)$ the object $K' = R\lim(K \otimes^{\mathbf{L}}_A K_n)$ is derived complete with respect to I.*

Proof. Since the category of derived complete objects is preserved under $R\lim$ it suffices to show that each $K \otimes^{\mathbf{L}}_A K_n$ is derived complete. By assumption for all $f \in I$ there is an e such that f^e is zero on $K \otimes^{\mathbf{L}}_A K_n$. Of course this implies that $T(K \otimes^{\mathbf{L}}_A K_n, f) = 0$ and we win. $\square$